ArbItro: Empirical Limits of Multi-Task CNN-RNN Pipelines for Football Foul Recognition

Carmine Gallo
University of Salerno
Department of Computer Science
Fisciano, Salerno, IT
c.gallo78@studenti.unisa.it

Andrea De Simone
University of Salerno
Department of Computer Science
Fisciano, Salerno, IT
a.desimone97@studenti.unisa.it

Matteo Trinchese
University of Salerno
Department of Computer Science
Fisciano, Salerno, IT
m.trinchese9@studenti.unisa.it

Salvatore Frontoso
University of Salerno
Department of Computer Science
Fisciano, Salerno, IT
s.frontoso@studenti.unisa.it

Salvatore Pio Ruggiero
University of Salerno
Department of Computer Science
Fisciano, Salerno, IT
s.ruggiero43@studenti.unisa.it

Abstract

Football foul recognition is a challenging video understanding problem due to severe class imbalance, subtle contact dynamics, and the need to interpret synchronized views under ambiguous officiating conditions. In this work, we present ArbItro, a multi-task CNN–RNN framework for football foul recognition on the SoccerNet-MVFouls benchmark, designed as a controlled study of what structurally classical architectures can recover under strong visual ambiguity and long-tail supervision. The proposed system decouples offence detection, action classification, and disciplinary severity assessment into separate prediction heads, with the aim of disentangling infringement recognition from sanction estimation. ArbItro combines two complementary architectures: a single-clip pipeline based on a CNN–BiLSTM backbone and a multi-clip pipeline that aggregates synchronized views through masked mean pooling. To address the long-tail distribution of the dataset, we employ targeted augmentation, focal-loss-based optimization, limited exploratory synthetic data injection, and task-specific loss weighting. Experimental results show that the ensemble achieves 46.87% balanced accuracy on the 3-class Severity task, while the binary Offence head remains affected by severe majority-class collapse. Rather than positioning ArbItro as a deployable decision-support system, we interpret these results as an empirical analysis of the limits of CNN–RNN architectures for automated refereeing under strong imbalance, occlusion, and visual ambiguity.

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CCS Concepts: • **Computing methodologies → Neural networks**.

Keywords: football, foul recognition, multi-view learning, multi-task learning, video understanding

1 Introduction

Professional football has evolved into a global industry where the financial ramifications of match outcomes extend far beyond the pitch. The economic stakes associated with officiating decisions are now critical, for instance, relegation from the English Premier League is estimated to result in a revenue decline of approximately £60–100 million in the first season alone, causing severe financial instability for the affected clubs [7]. Similarly, failing to qualify for the UEFA Champions League can represent a revenue shock of €20–60 million due to lost broadcasting distributions and prize money [7, 32]. In this high-stakes environment, the cost of errors, where incorrect refereeing decisions precipitate loss of sponsorship, investor confidence, or league status, has been identified as a significant source of economic damage and organizational instability [2]. Consequently, the demand for decision-making accuracy is driven not only by sporting integrity but by the imperatives of financial sustainability.

To address these challenges, the Video Assistant Referee (VAR) system was introduced to minimize "clear and obvious errors." While VAR has demonstrably improved decision accuracy, its current implementation has introduced significant friction regarding match fluidity and spectator experience [9]. Sociological and fan-sentiment analyses indicate that VAR-induced interruptions frequently disrupt the natural rhythm of the game, creating a "limbo" state that erodes emotional engagement and significantly reduces perceived match fluency [9]. Furthermore, the latency inherent in current review processes fosters anxiety and inhibits spontaneous goal celebrations, with fans reporting that the looming threat of video review dampens the immediate joy of the sport. The

trade-off between accuracy and the "flow of the game" remains a contentious issue, as the time required to reach a correct decision often outweighs the perceived benefit in fan satisfaction models [9].

The primary technical bottleneck in current VAR workflows lies not only in the human-in-the-loop dependency, which requires operators to manually review footage, select camera angles, and interpret complex visual evidence under time pressure, but also in the lack of computationally efficient video understanding systems capable of operating under strong temporal, semantic, and infrastructural constraints. While recent work in automated refereeing has shown the potential of increasingly expressive architectures, the field is progressively moving towards heavier spatiotemporal and multimodal models, including transformer-based video encoders and vision-language systems [1, 16, 31]. In this context, an important open question is not only how to maximize recognition accuracy, but also how far a lighter and more structured architecture can go when applied to highly imbalanced multi-view foul recognition.

In this work, we present ArbItro as a controlled empirical study of multi-view football foul recognition on SoccerNet-MVFouls. First, we introduce a decoupled multi-task formulation that separates binary offence detection, action classification, and disciplinary severity assessment, with the aim of clarifying whether disentangling infringement recognition from sanction estimation improves robustness under strong label imbalance [26]. Second, we propose a dual-pipeline design that combines a single-clip CNN–BiLSTM branch with a multi-clip masked-aggregation branch, allowing us to contrast broadcast-view stability with multi-view contextual aggregation. Third, rather than optimizing for parameter minimization, we intentionally adopt a structurally classical architecture, comprising 58.4M and 56.1M trainable parameters in its two pipelines, in order to assess how much disciplinary signal can be recovered under extreme class imbalance and visual ambiguity without relying on heavier spatiotemporal or multimodal encoders. Finally, through a detailed per-class evaluation, we show that this family of CNN–RNN pipelines remains strongly limited by majority-class collapse on offence recognition and by reduced robustness on rare and visually complex events, thereby motivating future work on richer spatiotemporal and multimodal representations [14–16].

2 Related Work

Automated analysis of sports videos is commonly built on top of large-scale human action recognition datasets and domain-specific benchmarks. Generic datasets such as Kinetics-400 and NTU RGB+D 120 have played a central role in pre-training visual backbones for video understanding [17, 22]. Kinetics-400 contains hundreds of thousands of short, unconstrained YouTube clips spanning 400 human action classes, and is widely used to initialize 2D and 3D convolutional networks for video classification [17]. NTU RGB+D 120 extends earlier RGB+D benchmarks with over 114k multimodal video samples across 120 action categories, providing synchronized RGB, depth, infrared, and 3D skeleton streams for view-invariant activity recognition and one-shot learning [22]. While these datasets offer broad coverage of everyday actions and human–object interactions, they do not capture the semantics and decision boundaries specific to football refereeing.

Within association football, the SoccerNet initiative has progressively established a suite of benchmarks targeting different aspects of broadcast analysis [3, 6, 12]. The original SoccerNet dataset focuses on action spotting in full-match broadcasts, providing temporal annotations for key events such as goals, substitutions, and cards [12]. Subsequent releases introduced additional tasks, including multi-view re-identification and tracking, dense captioning, camera calibration, and game state reconstruction [4, 5]. In this ecosystem, SoccerNet-MVFouls is the first benchmark dedicated specifically to foul recognition from synchronized multi-view clips [14, 29]. It contains 3,901 whistle-blown incidents extracted from professional matches, each comprising at least one main broadcast view and one or more replay views, annotated by an experienced referee with ten categorical attributes and a five-level severity scale [14]. This unique combination of multi-view footage and rich refereeing metadata makes SoccerNet-MVFouls the natural testbed for developing decision-support systems in VAR-like settings. A high-level comparison with other video action datasets is reported in Table 1, which highlights the unique multi-view and refereeing-oriented nature of SoccerNet-MVFouls.

Table 1. Comparison of relevant video action understanding datasets.

Dataset	Context	Task	View
Kinetics 400	Human actions	Classification	Single-view
NTU RGB+D 120	Human actions	Classification	Multi-view
SoccerNet-v2	Soccer	Classification	Single-view
SoccerNet-v3	Soccer	Re-identification	Multi-view
SoccerNet-MVFouls	**Soccer (Fouls)**	**Classification**	**Multi-view**

Beyond dataset design, a broad literature addresses high-level understanding of football broadcasts, including event spotting, tracking, and player-centric analytics. Transformer-based models such as ASTRA target fine-grained action spotting under long-tail distributions and label noise, demonstrating the benefits of temporal attention mechanisms and multi-modal inputs on SoccerNet benchmarks [34]. Recent works on tracking and player localization further exploit SoccerNet tracking and camera calibration challenges to improve multi-object tracking, pitch registration, and game

state reconstruction from moving broadcast cameras [4, 5]. In parallel, automated foul recognition and semi-automated VAR systems have been investigated using multi-view architectures and rich disciplinary annotations on SoccerNet-MVFouls [14–16]. Recent studies have also explored pose-based foul prediction from soccer broadcasts [10], while broader work on fine-grained action classification in sports highlights the importance of temporally precise motion modeling for tactical decision support [19]. A recent survey on deep learning for sports video event detection further contextualizes these developments within the wider sports analytics literature [35]. More recently, automated refereeing research has started to move beyond conventional task-specific video classifiers towards architectures with stronger semantic and multimodal reasoning capabilities. In particular, the evolution from VARS-style systems to explainable frameworks such as X-VARS highlights a broader shift from purely kinematic discrimination to multimodal representations enriched with language-aware reasoning [14–16]. At the same time, this shift typically increases architectural complexity and inference cost, especially when compared with lighter 2D CNN plus recurrent temporal aggregation pipelines. Within this broader design space, it is scientifically relevant to assess not only absolute recognition performance, but also which failure modes persist when foul recognition is approached through structurally classical architectures rather than through full spatiotemporal attention or multimodal reasoning. ArbItro is therefore positioned not as a state-of-the-art contender, nor merely as a generic baseline, but as a controlled reference study for characterizing the limits of multi-task CNN–RNN pipelines on SoccerNet-MVFouls. Its main scientific role is analytical: to assess whether explicit task decoupling, heterogeneous clip aggregation, and imbalance-aware optimization can recover meaningful disciplinary signal despite the severe occlusion, ambiguity, and long-tail effects that characterize the refereeing domain. A detailed quantitative comparison with prior VARS-based systems is provided in Section 6.5.

3 Problem Formulation

We formalize ArbItro as a supervised multi-task video classification framework. Given a set of multi-view video clips $C = \{c_1, c_2, \ldots, c_K\}$ associated with a single whistled incident, the system produces a tuple of predictions $(\hat{y}_{\text{off}}, \hat{y}_{\text{act}}, \hat{y}_{\text{sev}})$ corresponding to offence detection, action classification, and disciplinary severity assessment, respectively.

Task Definition. Each incident is annotated with three primary labels:

- **Offence** $y_{\text{off}} \in \{0, 1\}$: binary label indicating the presence or absence of a foul.
- **Action class** $y_{\text{act}} \in \{0, \ldots, 8\}$: categorical label encoding the type of physical intervention.
- **Severity** $y_{\text{sev}} \in \{0, 1, 2\}$: ordinal label representing the disciplinary sanction (No Card, Yellow Card, Red Card).

The three tasks are structurally related but exhibit distinct imbalance profiles and visual ambiguity levels, motivating a decoupled multi-task formulation rather than a single joint classifier.

Input Representation. Each video clip c_k is decoded into a fixed-length RGB tensor $\mathbf{X}_k \in \mathbb{R}^{T \times H \times W \times 3}$, where $T = 16$ frames, $H = 224$, $W = 398$. A scalar replay-speed value $s_k \in \mathbb{R}^+$ is retained as auxiliary contextual input. In the multi-clip setting, the system processes up to $K = 4$ clips per incident, applying masked mean pooling to accommodate variable numbers of available views.

Performance Measure. Given the severe long-tail distribution of the dataset, standard accuracy is an unreliable performance indicator. All primary heads are therefore evaluated using **Balanced Accuracy**, formally defined in Section 6.2.

4 Data Acquisition and Understanding

For the ArbItro project, we rely on the official dataset releases provided by SoccerNet and their accompanying public repositories [28, 29]. In particular, we adopt the SoccerNet-MVFouls benchmark as our sole source of supervision [14]. As discussed in Section 2 , SoccerNet-MVFouls occupies a unique position within the landscape of soccer video understanding datasets, providing synchronized multi-view clips and rich refereeing annotations tailored to foul recognition.

Furthermore, SoccerNet-MVFouls is the first dataset dedicated entirely to refereeing decisions, offering ten distinct manually annotated properties per action. This unique combination of synchronized multi-view footage and comprehensive, domain-specific metadata makes SoccerNet-MVFouls the natural testbed for studying the limits of multi-task video classification under realistic refereeing constraints.

4.1 Original Dataset Source

The dataset consists of 3,901 available actions, originally extracted as 5-second clips (starting 3 seconds before the action and ending 2 seconds after). Each action is composed of at least two video clips depicting the live action and at least one replay. We utilized the SoccerNet-MVFouls dataset [14], a specialized subset of the larger SoccerNet ecosystem, and we adopt its official split into a training set (2,916 actions), validation set (411 actions), test set (301 actions), and a *challenge* set (273 actions without annotations). Crucially, the foul-related properties are manually annotated by a professional soccer referee (6 years of experience and more than 300 official games), which strengthens the reliability of the ground truth used for training and evaluation. We did not use the *challenge* split as it lacks the required annotations.

Class Imbalance and Distribution. A critical characteristic of the dataset, as highlighted in the reference benchmark, is the significant long-tail distribution. Severe fouls are naturally rarer than minor infractions. As shown in Table 2, the "No Card" class dominates the severity distribution (55.3%), while "Red Card" events are extremely scarce (less than 2%). Similarly, the action types are heavily skewed towards "Tackling" and "Standing Tackling", which combined account for nearly 60% of the dataset.

Original Label Space. The annotations expose three primary targets used as the main prediction heads in our architecture:

- **Offence**: {No offence, Between, Offence}.
- **Action class**: A comprehensive set of action types including {Tackling, Standing tackling, High leg, Holding, Pushing, Elbowing, Dive, Challenge, Don't know}.
- **Severity**: A 5-level scale representing the disciplinary sanction intensity.

In addition, the dataset provides several auxiliary attributes that enrich the description of the incident and are used to train the regularization heads:

- **Touch ball**: {No, Maybe, Yes}.
- **Try to play**: {No, Yes}.
- **Contact**: {Without contact, With contact}.
- **Bodypart**: {Under body, Upper body}.
- **Handball**: {No handball, Handball}.

Table 2. Original dataset distribution (SoccerNet-MVFouls).

Category	Class	Freq. (%)
Offence	Offence	85.6%
	No offence	11.1%
	Between	3.3%
Severity	No Card	48.0%
	Yellow Card	23.5%
	NC/YC (Borderline)	13.8%
	Empty	12.1%
	YC/RD (Borderline)	1.5%
	Red Card	1.1%
Action Type	Standing tackling	43.3%
	Tackling	15.5%
	Challenge	13.1%
	Holding	12.4%
	Elbowing	6.1%
	High leg	3.5%
	Pushing	3.0%
	Don't know/Unknown	2.2%
	Dive	1.0%

Overall, this structure supports our multi-task formulation: a single action instance is described by multiple categorical fields, allowing the model to learn correlated features.

4.2 Data Cleaning and Mapping

As a first step, we performed a cleaning and mapping procedure over the original annotations to (i) reduce noise due to missing or inconsistent strings and (ii) obtain compact, learnable target spaces for both the main heads and the auxiliary attributes.

4.2.1 Preprocessing Steps. To ensure consistency, we applied a small set of manual corrections to resolve syntactic inconsistencies in the raw annotations (e.g., "Shoulder" vs. "Shoulders") in order to avoid spurious label duplication.

We redefined the label space for our primary (*Offence*, *Severity*, and *Action class*) and auxiliary classification heads to facilitate training, as detailed in Table 3 and Table 4, respectively.

Table 3. Label mapping strategy for primary multi-task heads.

Task Head	Original Annotations	Mapped Value
Offence	Empty, Between, No Offence	0
	Offence	1
Severity	1.0, 2.0 (No Card)	0
	3.0, 4.0 (Yellow Card)	1
	5.0 (Red Card)	2
Action	Empty, Don't know	0
	Challenge	1
	Tackling	2
	Standing tackling	3
	High leg	4
	Holding	5
	Pushing	6
	Elbowing	7
	Dive	8

After target processing and label unification, the resulting class distribution is reported in Figure 1.

Clip-level Processing. For each clip, we ensured the corresponding video file path was valid by appending the expected extension (.mp4) and verifying file existence. Each clip was then decoded into a fixed-size video tensor with shape $(\text{Batch}, 16, 224, 398, 3)$, strictly adhering to the aspect ratio and temporal depth defined in the SoccerNet-MVFouls benchmark, as illustrated in Figure 2. The Replay speed field was retained as a scalar float feature (e.g., 1.0, 2.8). The Camera type string was mapped into three discrete categories: 0 for *Main* (wide view), 1 for *Close-up* (zoom/detail), and 2 for *Special* (e.g., spider/goal cameras). The Timestamp was not used in our training pipeline.

4.3 Class Balancing Strategies

The distribution of offences in football is naturally long-tailed, with "No Offence" or minor fouls being far more

Table 4. Label mapping strategy for auxiliary multi-task heads.

Task Head	Original Annotations	Mapped Value
Contact	Without contact, Empty	0
	With contact	1
Touch ball	No, Empty	0.0
	Maybe	0.5
	Yes	1.0
Try to play	No (aggressive)	0.0
	Empty (neutral/unknown)	0.5
	Yes (fair attempt)	1.0
Multiple fouls	No, Empty	0
	Yes	1
Bodypart	Empty (None)	0
	Under body	1
	Upper body	2
Upper body part	Empty	0
	Arms	1
	Shoulder	2
Handball	No handball, Empty	0
	Handball	1

frequent than severe "Red Card" offences. We address this imbalance through a two-fold strategy: severity-aware oversampling and loss weighting.

4.3.1 Augmentation Strategies and Synthetic Data.

To address the complexity of the action recognition task and, more critically, the extreme scarcity of severe foul examples (i.e., "Red Card") in the original distribution, we employed a hybrid strategy combining algorithmic data augmentation and the injection of synthetic data.

During training, we apply a systematic and predefined augmentation pipeline utilizing the OpenCV library, targeting specifically the minority classes to prevent overfitting on the few available severe examples. While the exact tensor specifications and scaling factors are detailed in the respective pipeline sections (5.2, 5.3), the core augmentation strategy relies on generating synthetic variations through a fixed set of isolated and complex transformations:

- **Spatial transformations**: Horizontal flipping, small-angle rotations, and zooming operations (both zoom-in and zoom-out) to simulate different camera distances and angles.
- **Photometric transformations**: Adjustments to brightness and contrast levels to simulate varying stadium lighting conditions.

Crucially, these transformations are applied consistently across all frames of a given video sequence to preserve the temporal coherence of the player kinematics.

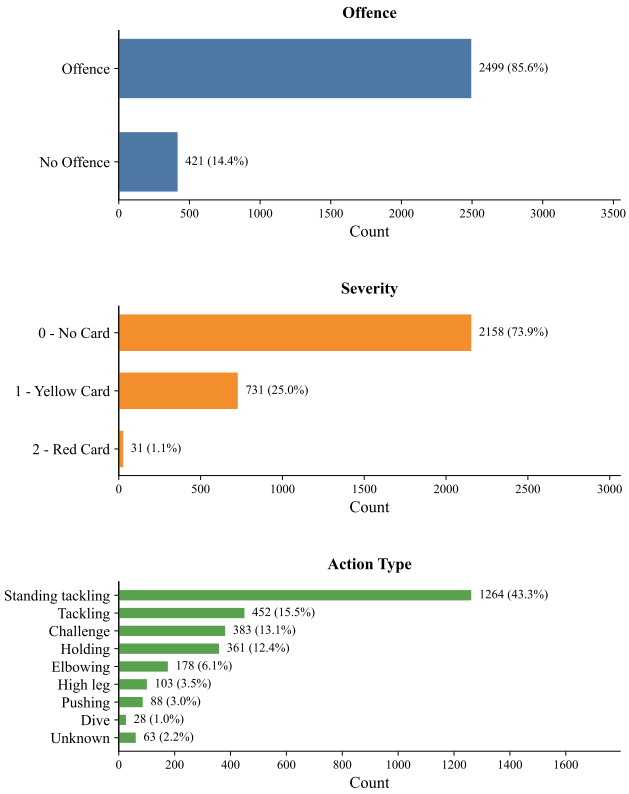

Figure 1. Class distribution of the training set after label processing.

Figure 2. Example replay clip represented as a sequence of 16 frames after temporal sampling and decoding.

Synthetic Data Injection. To further investigate possible mitigation strategies for extreme class imbalance, we conducted an exploratory proof-of-concept using a highly restricted set of synthetic clips extracted from the football video game *FIFA 23*, a strategy supported by foundational literature on extracting ground truth from commercial game engines [25]. Specifically, we generated 4 synthetic clips for

the training stage and 10 additional clips for auxiliary validation analysis, targeting the rarest outcome in our primary label space, namely the "Red Card" class for the Severity head. These clips were obtained by recording replay sequences from multiple virtual camera angles and were manually annotated to approximate the semantics of severe foul incidents. Given the very limited scale of this synthetic subset, we do not treat it as evidence of statistically robust generalization gains; rather, we use it as a qualitative probe to assess whether synthetic replay data may expose the model to rare occlusion patterns and severe-contact configurations that are scarcely represented in the original benchmark.

Severity-Aware Oversampling. Instead of applying data augmentation uniformly, we implemented a dynamic sampling strategy within the data generator that inversely correlates with class frequency. Based on the *Severity* label, samples belonging to minority classes are systematically augmented with higher intensities compared to those in the majority classes. This approach explicitly forces the model to process the rarest and most critical events (e.g., red card) significantly more often during an epoch, counteracting the natural bias of the dataset. The specific oversampling factors and augmentation configurations are detailed in the respective pipeline sections (5.2, 5.3).

Loss Function and Task Weighting. At the optimization level, we address the class imbalance by employing specialized objective functions, such as Focal Loss, specifically targeted at the most imbalanced prediction heads. This approach down-weights easy negatives and focuses the learning signal on hard, misclassified examples. Additionally, within our multi-task formulation, the individual loss components are combined using a weighted sum. By assigning higher scalar weights to the most critical tasks (such as severity assessment) compared to standard or auxiliary heads, we prioritize the correct classification of disciplinary sanctions. The exact hyperparameters for the loss functions and the task weights are optimized and reported individually for each pipeline.

5 Methodology and Architecture

In this section, we present the architectural choices adopted to study the behavior of lightweight multi-task models for multi-view football foul recognition. Rather than targeting a production-ready refereeing system, the proposed design is intentionally used as a controlled experimental setting in which task decoupling, multi-clip aggregation, and imbalance-aware optimization can be evaluated under realistic broadcast constraints.

The core system processes sequences of RGB frames through a Convolutional Neural Network (CNN) backbone for spatial feature extraction, followed by a Recurrent Neural Network (RNN) for temporal aggregation. This hierarchical structure allows the model to capture both the fine-grained details of a contact (e.g., body part involvement) and the dynamic evolution of the action (e.g., impact momentum).

We first outline the general hardware and software infrastructure supporting the system. Subsequently, we detail the primary processing pipeline in Section 5.2 and discuss the secondary architecture in Section 5.3. We then present an ensemble approach that combines the strengths of both pipelines, and finally describe the end-user interface designed to interpret and visualize the model's predictions.

5.1 General System Architecture

The ArbItro framework is designed as a modular pipeline that bridges raw multi-view video ingestion and structured multi-task prediction. Its purpose is not to emulate a fully autonomous officiating process, but to provide a controlled architectural setting in which the interaction between spatial encoding, temporal aggregation, and multi-task supervision can be systematically evaluated.

End-to-End Workflow. The system operates as an end-to-end pipeline. Raw video files are ingested and preprocessed to match the tensor requirements of the neural network. These tensors are then processed through the deep learning model, and the output probability distributions are finally mapped to interpretable refereeing decisions displayed via the user interface, described in Section 5.4.

Software Stack. The core logic is implemented in Python using the TensorFlow/Keras framework for model definition and training, and is exposed through a lightweight inference server that provides asynchronous prediction endpoints.

Visual Feature Extraction (Spatial). We employ a Time-Distributed InceptionResNetV2 backbone [30] for frame-wise spatial feature extraction. To leverage transfer learning, the model is pre-trained on ImageNet [8]. A "partial freeze" strategy is applied: the first 70% of the layers are frozen to retain generic feature extraction capabilities, while the top 30% are fine-tuned to learn football-specific patterns. The frame-wise feature maps are reduced via a GlobalAveragePooling2D layer and regularized with Dropout.

The choice of InceptionResNetV2 should be interpreted as a controlled architectural decision rather than as a claim of efficiency by contemporary standards. Given the limited scale and strong imbalance of SoccerNet-MVFouls, a mature convolutional backbone provides a reproducible feature extraction regime in which the effects of task decomposition and multi-view aggregation can be studied without conflating them with the optimization instability of more data-hungry architectures.

Temporal Modeling (Sequential). To aggregate the sequence of spatial frame features into a single temporal action descriptor, we utilize a Bidirectional LSTM (BiLSTM) [27]

with 256 units. The bidirectional approach allows the network to process the foul context in both forward and reverse time, capturing the causal relationship between the approach, the contact, and the aftermath.

5.2 First Pipeline

The first pipeline, which evaluates the action using a single television broadcast clip, is designed to capture both the spatial details of a foul (e.g., limb positions) and the temporal dynamics (e.g., speed of impact) by fusing visual data with metadata.

Input Preprocessing. To address class imbalance, a strategic data augmentation pipeline is implemented within the data loader. The original base sample is always preserved in the dataset. When augmentation is enabled, supplementary samples are generated following a strict priority-based heuristic. The highest priority is assigned to severe offenses ("Red Card", corresponding to severity class 2), which are extensively augmented to produce a total of ten variants per original sample. The nine transformations applied include horizontal flipping, minor rotations, a combination of flipping and rotation, zooming in and out, brightness and contrast modifications, as well as composite transformations (rotation with contrast, and zoom with brightness). Secondly, intermediate offenses ("Yellow Card", severity class 1) receive a single augmentation via horizontal flipping, effectively doubling their representation.

Context Fusion and Multi-Task Heads. Following the extraction of spatiotemporal features from the single television broadcast clip via the BiLSTM, the architecture integrates additional contextual information. To capture the dynamic context of the foul, the recurrent video embedding is fused with the action's replay speed. This scalar speed input is processed independently through a dedicated fully connected layer (32 units, ReLU activation) and regularized via Dropout ($p = 0.3$). The resulting representations from the video and speed branches are then concatenated into a single, holistic feature vector. This combined representation is projected into a shared latent space through a robust dense block of 512 units. To promote training stability and generalization, this shared backbone incorporates Batch Normalization and structural regularization (Dropout with $p = 0.4$), serving as the informational foundation for all subsequent multi-task decisions. To model the complex and multifaceted nature of refereeing decisions, the architecture branches this shared representation into multiple classification and regression heads, categorized into primary and auxiliary tasks. For the primary tasks, specific regularization strategies are applied to intermediate dense layers before the final predictions: the Severity head utilizes Dropout ($p = 0.3$) and L2 regularization ($\lambda = 0.02$), the Action head employs heavier Dropout ($p = 0.55$) and L2 regularization ($\lambda = 0.01$), while the binary Offence head connects directly to the shared backbone without additional intermediate regularization.

Multi-Task Outputs. Building on the branching strategy described above, the model outputs three primary refereeing decisions: *Offence Prediction* (a binary classification via sigmoid activation to determine the presence or absence of an infringement), *Action Classification* (a 9-class softmax output categorizing the specific type of intervention, e.g., Challenge, High leg, Dive), and *Severity Assessment* (a 3-class softmax output evaluating the gravity of the foul to predict the referee's card decision). Simultaneously, to encourage the shared backbone to learn fundamental physical dynamics, five auxiliary heads are connected directly to the shared layer. These act as semantic regularizers: presence of physical contact (binary classification), body part involved (3-class classification), actual contact with the ball (continuous regression), possible handball (binary classification), and intention or attempt to play the ball (continuous regression). The proposed architecture is shown in Figure 3.

Loss Formulation and Class Imbalance. The architecture is trained end-to-end by minimizing a linear combination of the cost functions from each head. To address the inherent class imbalance of refereeing events, where severe infractions are significantly less frequent than regular play, the critical Severity head eschews standard Cross-Entropy in favor of a custom Focal Loss [21] ($\gamma = 2.5$, $\alpha = 0.25$). This dynamically scales the gradient based on prediction confidence, forcing the model to focus on hard-to-classify examples, while the remaining primary heads (Offence and Action) rely on standard binary and categorical Cross-Entropy, respectively.

Finally, the overall objective function is weighted to reflect the hierarchical priority of the predictions. Primary tasks dominate the gradient with a weight of $\lambda = 8.0$ for Severity, $\lambda = 4.0$ for Action, and $\lambda = 3.0$ for Offence. The auxiliary metrics contribute as soft regularizers with a minimal weight of $\lambda = 0.1$ each. This ensures that the auxiliary tasks guide feature extraction without overpowering the primary optimization objectives.

5.3 Second Pipeline

The second pipeline was developed to perform a multi-clip analysis (ranging from 2 to 4 clips) for each sample in the dataset. The focus was on analyzing the action type and evaluating the Offence/No Offence label. For the multi-clip evaluation, a masked mean approach was introduced: the model evaluates the clips individually and then outputs a single prediction based on the average of these multiple evaluations. We did not implement a camera-weighted average, as it decreased the model's accuracy; specifically, many "No Card" instances were misclassified as "Red Card".

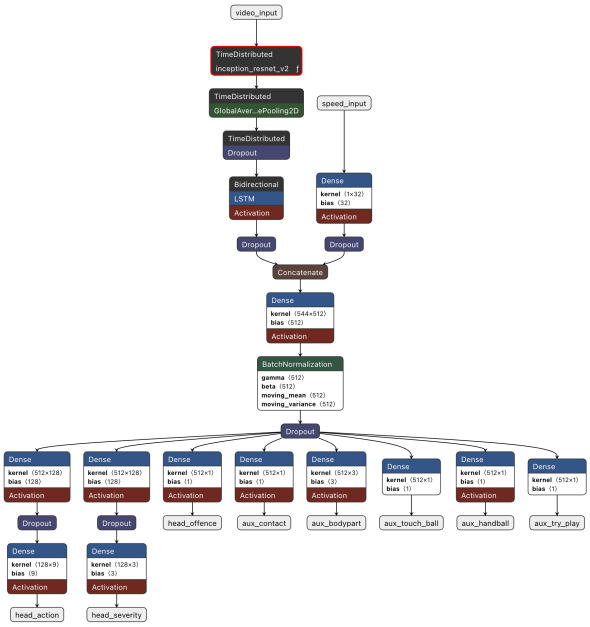

Figure 3. Architecture of the first pipeline.

Input preprocessing. To further enhance the robustness of fine-grained action recognition, the second pipeline introduces a targeted data augmentation strategy for underrepresented classes. Specifically, instances labeled with predefined rare actions (i.e., "High leg", "Elbowing", "Pushing", "Dive", and "Holding") are subjected to a single horizontal flip, provided they have not already been augmented under the previously described severity conditions.

Furthermore, as illustrated in Figure 4, the adopted augmentation strategy partially mitigates the imbalance affecting the most underrepresented target classes, particularly for the primary refereeing outcomes. However, in the proposed multi-task setting, each augmented sample simultaneously contributes labels to all prediction heads. Consequently, rebalancing one task by replicating selected samples may also alter the class distribution of the remaining heads, in some cases increasing their relative imbalance. For this reason, a more aggressive balancing strategy was not adopted, as it would have risked over-amplifying a limited subset of correlated label combinations and further distorting the overall data distribution. The final augmentation policy should therefore be interpreted as a controlled compromise between improving the representation of rare classes and preserving a sufficiently coherent multi-task label structure.

Multi-Clip Aggregation and Masking. Unlike the single-clip approach, this model evaluates multiple video clips for each action sample. The overarching feature representation is derived through a masked mean pooling operation applied to the individual clip-level embeddings. To aggregate the

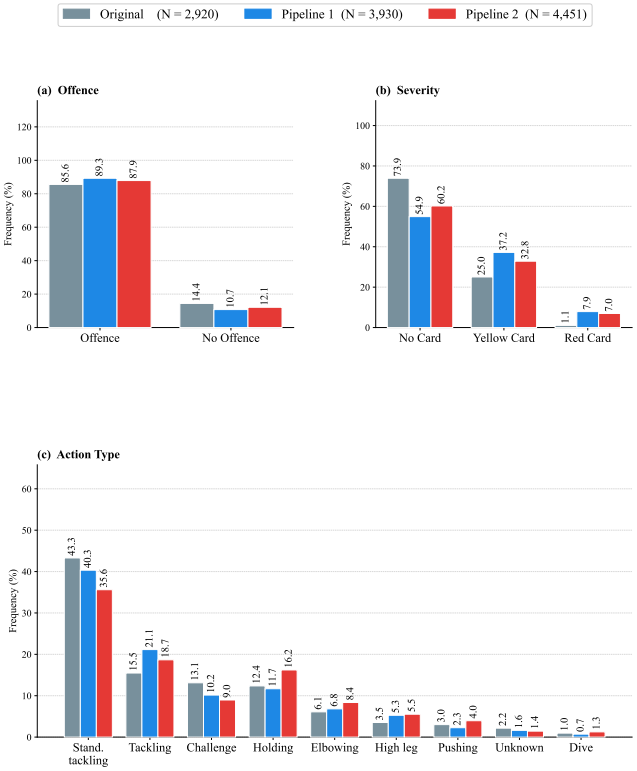

Figure 4. Class distributions across all prediction heads for the original dataset and the augmented datasets used in first and second pipeline.

temporal dynamics into a single representation, we apply a masked mean pooling operation over the clip-level embeddings. Given a maximum of K clips per action ($K = 4$), let $h_k \in \mathbb{R}^D$ be the feature vector extracted from the k-th clip. We define a binary mask vector $m \in \{0, 1\}^K$, where $m_k = 1$ if the clip contains valid visual information and $m_k = 0$ if it is padding. The final multi-view tensor V_{final} is computed as:

$$V_{final} = \frac{1}{\sum_{k=1}^{K} m_k} \sum_{k=1}^{K} (m_k \cdot h_k) \tag{1}$$

This mathematical formulation ensures that padded clips receive a weight of zero, completely excluding them from the final aggregation without skewing the temporal average.

Context Fusion and Multi-Task Heads. Building upon the fusion strategy established in the first pipeline, this secondary architecture adapts the shared backbone to effectively handle multi-clip inputs. Following the extraction of clip-level features, the temporal dynamics are aggregated into a single, stable video embedding via the aforementioned masked mean pooling. This distilled representation is then concatenated with the game dynamics (the average replay

speed), which is processed through the same 32-unit dedicated dense branch used in the primary pipeline, but with a slightly reduced Dropout rate ($p = 0.20$).

Crucially, to accommodate the richer representation provided by the multi-clip aggregation while mitigating the increased risk of overfitting, we reduced the capacity of the shared latent space from 512 to 256 units. Because the masked mean pooling operation inherently filters and stabilizes the spatiotemporal information across the sequence, this 256-unit bottleneck forces the network to retain only the most salient, cross-clip features without memorizing noise. Furthermore, to robustly handle the variable-length nature of the multi-clip inputs and enforce stricter regularization on this compact representation, we replaced Batch Normalization with Layer Normalization and introduced an L2 weight penalty ($\lambda = 0.001$) alongside a slightly higher Dropout ($p = 0.45$). This refined, highly regularized backbone then diverges into the multi-task heads. Unlike the primary pipeline, which applied custom regularization to each specific head, this architecture standardizes the primary decision heads (Severity, Action, and Offence): each is preceded by an intermediate dense layer uniformly regularized with L2 ($\lambda = 0.005$) and Dropout ($p = 0.40$) to ensure balanced learning and prevent negative transfer across tasks.

Multi-Task Outputs. The second pipeline maintains the identical multi-task output structure established in the primary architecture, deriving three main refereeing decisions and five auxiliary semantic regularizers from the shared backbone. However, to improve learning stability and address class distribution, the *Action Classification* target was restructured during the training and validation phases: the original 9 fine-grained action types were aggregated into 4 strategic macro-classes (e.g., neutral/no impact, standard tackles, upper-body fouls, and dangerous play), and the corresponding softmax output was adjusted to 4 units. The overall architecture of the second pipeline is illustrated in Figure 5. The first pipeline comprises 58.4 million trainable parameters, whereas the second pipeline comprises 56.1 million trainable parameters. These values are not negligible in absolute terms and should not be described as lightweight by the standards of contemporary efficient video architectures. In our setting, this parameter budget is primarily a consequence of the adopted ImageNet-pretrained InceptionResNetV2 backbone rather than an explicit attempt to minimize model size. The scientific role of this design is therefore different: the framework serves as a controlled reference architecture for testing how much disciplinary information can be recovered through 2D spatial encoding and recurrent temporal aggregation alone, before introducing heavier 3D convolutional, transformer-based, or multimodal mechanisms.

Loss Formulation and Class Imbalance. The architecture is trained end-to-end by minimizing a linear combination of the cost functions from each head. To complement

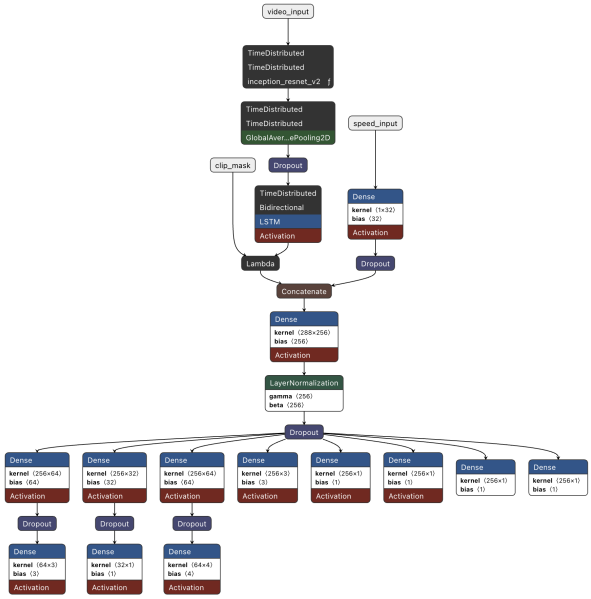

Figure 5. Architecture of the second pipeline.

the data augmentation strategy and further address the severe class imbalance typical of refereeing events, the primary heads eschew standard Cross-Entropy. Instead, we deployed custom implementations of the Focal Loss (both binary and multiclass) with baked-in class weights. The Focal Loss dynamically scales the gradient based on prediction confidence, forcing the model to focus on hard examples, while applying explicit balancing factors. Specifically, the Severity head employs a weighted Focal Loss with $\gamma = 2.0$ and a manual linear scale of class-specific multipliers (weights of $[1.0, 1.5, 2.0]$) to penalize misclassifications of rarer cards. Similarly, the Action head utilizes $\gamma = 2.0$ with targeted manual weights ($[1.0, 1.0, 1.5, 3.0]$), and the binary Offence head uses a weighted binary Focal Loss with $\gamma = 2.0$ and a specific penalty weight of 2.0 assigned to the rarer "No Offence" class.

Finally, the overall objective function is weighted to reflect the hierarchical priority of the decisions. The primary tasks dominate the gradient, with a weight of $\lambda = 7.0$ assigned to both Offence and Action, and $\lambda = 4.0$ to Severity. The auxiliary metrics contribute to a lesser extent to ensure they guide feature extraction without overpowering the main optimization objectives: the physical contact loss is weighted at $\lambda = 2.0$, while the remaining four auxiliary tasks receive a weight of $\lambda = 0.5$ each.

5.4 The ArbItro Interface

To make the model's predictions accessible to referees and analysts, we developed a local graphical interface that abstracts the underlying tensor operations.

Technology Stack. The interface is built using a client-server architecture:

- **Backend:** A Python server hosts the trained model. It exposes endpoints to receive video files and return JSON-formatted predictions.
- **Frontend:** The visual dashboard is implemented using standard web technologies (HTML5, CSS3, JavaScript). This ensures a lightweight, responsive user experience without the need for heavy compiled frameworks.

Inference Flow. When a user uploads one or more videos through the interface, the backend intercepts the files and triggers the preprocessing and inference pipeline. For each action instance, the system accepts up to four videos, which are converted into tensors and forwarded to the loaded ensemble model. The resulting probability distributions are then returned to the frontend, where the JavaScript layer visualizes the predicted severity, offence decision, and foul category together with their associated confidence scores. As an indicative measure of practical responsiveness, we evaluated the end-to-end response time on a local ARM-based Apple Silicon M2 system with 16 GB of unified memory. The first request required 150 seconds, due to model initialization and runtime warm-up overhead, whereas subsequent requests showed a lower average latency of approximately 11 seconds. Although these measurements do not support real-time deployment, they provide a practical reference point for interpreting the computational footprint of the proposed framework. In this paper, latency is therefore reported not as evidence of operational readiness, but as contextual information for situating the architectural simplicity of the system relative to its still-limited predictive reliability.

6 Experiments and Evaluation

In this section, we present the empirical evaluation of the ArbItro system. First, we detail the hardware infrastructure and the training configuration adopted to develop the models and ensure reproducibility. Second, we formalize the evaluation metrics, which have been specifically selected to address the severe class imbalance inherent in the soccer refereeing domain. Finally, we report quantitative results and provide a qualitative analysis of both pipelines and their ensemble, including a comparison with prior refereeing-oriented systems. Rather than emphasizing only absolute performance, the analysis focuses on which disciplinary signals remain recoverable and which failure modes persist under strong imbalance and visual ambiguity.

6.1 Experimental Setup

To ensure reproducibility and manage the high computational cost of video processing, we used a cloud-based high-performance computing environment. In the interest of transparency and to encourage follow-up work, we publicly release ArbItro's complete source code, training pipelines, and graphical user interface at https://github.com/gallocarmine/ArbItro.

Hardware Infrastructure. Training and validation were performed on the **Google Colab Pro+** platform, using a total budget of 300 compute units (134 hours of runtime). The specific hardware configuration included:

- **GPU:** NVIDIA L4 Tensor Core GPU (22.5 GB VRAM), and NVIDIA A100 (40 GB VRAM) utilized for mixed-precision training.
- **System Memory:** 53.0 GB (L4) and 80 GB (A100) of High-RAM.
- **Storage:** A persistent cloud storage (Google Drive) mounted directly into the runtime environment to stream the SoccerNet-MVFouls dataset without local caching bottlenecks.

Training Configuration. The model was trained using the Adam optimizer [18] with an initial learning rate of 1.0×10^{-4}. To maximize GPU hardware utilization while strictly avoiding Out-Of-Memory (OOM) bottlenecks, the batch size was tailored to the memory footprint of the respective architectures. Specifically, we employed a batch size of $B = 32$ sequences for the single-clip primary pipeline, whereas the batch size was reduced to $B = 8$ for the computationally heavier multi-clip secondary pipeline. To further optimize convergence, a dynamic learning rate scheduler (ReduceLROnPlateau) was uniformly applied across both pipelines. This mechanism monitored the target loss, decaying the learning rate by a factor of 0.5 upon a plateau of 6 epochs, down to a minimum threshold of 1.0×10^{-7}. Finally, the training procedure incorporated a robust suite of callbacks: an Early Stopping mechanism evaluating the validation Balanced Accuracy to prevent overfitting, alongside a model checkpointing system designed to systematically save the optimal network weights achieved during the best-performing epochs.

6.2 Evaluation Metrics

Consistent with the performance objectives defined in Section 3, we employ metrics specifically designed for imbalanced domains. Standard accuracy is discarded as a primary metric, as a trivial classifier predicting "No Card" would achieve more than 55% accuracy while failing the system's core purpose.

Binary Balanced Accuracy. For the binary classification task, Offence head, we define the Balanced Accuracy as the arithmetic mean of Sensitivity (True Positive Rate) and Specificity (True Negative Rate). Given the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), the metric is defined as:

$$\text{BalAcc}_{bin} = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right) \tag{2}$$

This ensures that the model is penalized equally for missing an offence (FN) and for hallucinating one (FP).

Multiclass Balanced Accuracy. For the multi-class tasks, Severity and Action head, we extend this concept by computing the macro-average of the recall for each class. Let C be the number of classes (e.g., $C = 3$ for Severity), and let M_{ij} be the entries of the confusion matrix where i represents the ground truth and j the prediction. The recall for a specific class i, denoted as R_i, is the fraction of correctly classified instances over the total instances of that class:

$$R_i = \frac{M_{ii}}{\sum_{j=1}^{C} M_{ij}} \tag{3}$$

The final Multiclass Balanced Accuracy is the mean of these per-class recalls:

$$\text{BalAcc}_{multi} = \frac{1}{C} \sum_{i=1}^{C} R_i \tag{4}$$

This formulation prevents the majority class (e.g., "No Card" or "Standing Tackling") from dominating the evaluation score.

6.3 First Pipeline Results

In this section, we present the evaluation of the first pipeline, which, in contrast to the subsequent multi-clip architecture, operates on a single video clip per action. The results, summarized in Table 5, demonstrate the immense challenge of fine-grained action recognition in highly imbalanced domains.

Table 5. First pipeline performance metrics.

Task Head	Accuracy	Balanced Accuracy
Severity	62.13%	**40.71%**
Offence	87.71%	50.00%
Action	35.55%	10.66%

The primary objective of ArbItro is to distinguish between disciplinary sanctions. The model achieved a raw accuracy of 62.13%, but the Balanced Accuracy fell to just 40.71%. Most notably, the model completely failed to identify the minority class "Red Card", yielding a recall of 0.00% (failing to identify any of the 5 instances), see Table 6.

Contrary to our initial hypothesis, the combination of Focal Loss and aggressive oversampling was insufficient to prevent the model from collapsing. Instead of learning the nuanced dynamics of severe fouls, the network heavily favored the majority classes, predicting "No Card" (65.18% recall) and "Yellow Card" (56.94% recall) while entirely ignoring the rarest and most critical events.

This mode collapse, defined as the tendency of the network to predict only the majority class, observed in the severity assessment, was equally evident in the other primary heads due to visual ambiguity:

Table 6. First pipeline: detailed confusion metrics for severity head.

Severity Class	Correct / Total	Class Recall
No Card	146 / 224	65.18%
Yellow Card	41 / 72	56.94%
Red Card	0 / 5	0.00%
Overall Metrics	**Acc:** 62.13%	**Bal. Acc:** 40.71%

For the *Offence* head, despite a high raw accuracy of 87.71%, the Balanced Accuracy remained exactly at 50.00%. The model predicted "Offence" for all 264 positive samples (100% recall) but failed completely on the 37 "No Offence" scenarios (0.00% recall).

For the *Action* head, the model struggled with fine-grained distinctions, achieving a Balanced Accuracy of only 10.66%. It gravitated heavily towards the dominant "Standing tackling" class (correctly identifying 98 out of 127 instances, or 77.17%), while failing entirely (0.00%) on minority actions like "Challenge", "High leg", and upper-body fouls, as shown in the Table 7.

Table 7. First pipeline: action recognition across all classes.

Action Class	Correct / Total	Class Recall
Standing tackling	98 / 127	77.17%
Tackling	9 / 48	18.75%
Challenge	0 / 50	0.00%
Holding	0 / 33	0.00%
Elbowing	0 / 21	0.00%
High leg	0 / 9	0.00%
Pushing	0 / 7	0.00%
Dive	0 / 5	0.00%
Unknown	0 / 1	0.00%
Overall Metrics	**Acc:** 35.55%	**Bal. Acc:** 10.66%

These results suggest that global spatiotemporal features extracted from a single clip are insufficient for resolving visual ambiguity in complex or rare refereeing events. More importantly, they indicate that the limiting factor is not merely class frequency, but the inability of global pooled frame descriptors to preserve the localized contact information required to distinguish subtle disciplinary boundaries.

6.4 Second Pipeline Results

In this section, we present the empirical evaluation of the second operational pipeline. This iteration introduces a multi-clip analysis strategy, utilizing a masked mean approach to aggregate predictions from multiple clips representing the same action.

As detailed in Table 8, overall, while the multi-clip approach increased the raw accuracy for certain tasks, it introduced a significant bias toward majority classes.

Table 8. Second pipeline performance metrics.

Task Head	Accuracy	Balanced Accuracy
Severity	72.67%	39.10%
Offence	92.67%	50.00%
Action	32.00%	17.83%

In particular, the multi-clip aggregation structurally altered the severity predictions. As detailed in Table 9, the model achieved a remarkably high accuracy on the background class ("No Card"). However, this came at the severe cost of minority recall, leading to an overall drop in Balanced Accuracy. The model exhibited a clear mode collapse, entirely failing to recognize intermediate fouls ("Yellow Card"). Despite this limitation, the model achieved a crucial improvement in detecting the most extreme infractions: the recognition of "Red Card" events increased from a complete failure in the primary pipeline to a 20% recall in this secondary iteration. It must be noted that the performance observed on the "Red Card" class is inherently limited by its severe underrepresentation in the official test split, which contains only 5 samples. Consequently, improvements on this class should be interpreted as qualitative indicators of sensitivity to extreme fouls rather than as statistically robust milestones.

Table 9. Second pipeline: detailed confusion metrics for severity head.

Severity Class	Correct / Total	Class Recall
No Card	217 / 223	97.31%
Yellow Card	0 / 72	0.00%
Red Card	1 / 5	20.00%
Overall Metrics	**Acc:** 72.67%	**Bal. Acc:** 39.10%

† One No Card instance was skipped during inference due to a batch-boundary drop in the test loader.

For the binary *Offence* task, the model remained entirely biased. It predicted the presence of a foul for every single instance in the test set, achieving perfect recall on positive instances but failing completely to identify "No Offence" scenarios.

Conversely, the *Action* classification demonstrated a tangible benefit from the multi-clip strategy and from the rare actions augmentation. The model learned to better differentiate among various types of tackles, distributing its predictions more evenly, and yielding a relative improvement of 67.26% in balanced accuracy compared to the first pipeline, as shown in Table 10. This redistribution is further detailed

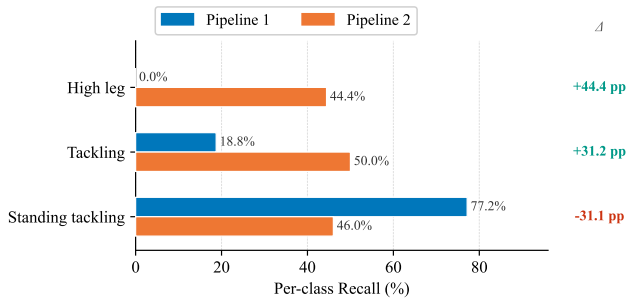

Figure 6. Per-class recall comparison on the Action head between first and second pipeline.

at the per-class level in Figure 6: while "Tackling" (+31.3 pp) and "High leg" (+44.4 pp) benefit most from the temporal aggregation across multiple viewpoints, "Standing tackling" regresses by −31.1 pp, suggesting that the model reallocates its representational capacity towards previously neglected minority classes rather than improving uniformly.

Table 10. Second pipeline: action recognition across all classes.

Action Class	Correct / Total	Class Recall
Tackling	24 / 48	50.00%
Standing tackling	58 / 126	46.03%
High leg	4 / 9	44.44%
Challenge	10 / 50	20.00%
Unknown	0 / 1	0.00%
Holding	0 / 33	0.00%
Pushing	0 / 7	0.00%
Elbowing	0 / 21	0.00%
Dive	0 / 5	0.00%
Overall Metrics	**Acc:** 32.00%	**Bal. Acc:** 17.83%

Furthermore, an interesting correlation emerges between the specific action classes more accurately recognized by the second pipeline (such as tackling variants and high legs) and the occurrence of severe infractions. However, successfully identifying the action type proves insufficient for the reliable prediction of a "Red Card". Statistically, these interventions predominantly correspond to "No Card" or "Yellow Card" outcomes in the ground truth. Because the multi-clip architecture suffers from a severe bias towards the majority "No Card" class, entirely failing to detect "Yellow Card" instances, the model systematically underestimates the disciplinary outcome. Consequently, the potential benefit of improved action recognition is largely neutralized by the background class bias, limiting the overall assignment of red cards.

6.4.1 Ensemble Strategy. To maximize the strengths of both individual approaches, we developed a hybrid ensemble

architecture combining the first and the second pipeline. In this configuration, the multi-clip model (second pipeline) serves as the primary decision-maker for the Offence and Action classification tasks.

For the Severity assessment, a specialized cascading logic is employed to refine the classification of disciplinary sanctions by combining the distinct input modalities of the two models. Within this ensemble framework, the second pipeline, leveraging its multi-clip analysis, acts as the initial evaluator, specifically tasked with the overarching detection of severe, high-impact infractions (i.e., "Red Card"). If the second pipeline does not predict a "Red Card", the decision is deferred to the first pipeline, which analyzes exclusively the single television broadcast clip. This primary model is then configured to decide solely between the "Yellow Card" and "No Card" classes by normalizing its corresponding output probabilities. This synergistic mechanism leverages the second pipeline's comprehensive multi-view sensitivity to extreme events, while relying on the first pipeline's focused stability on the main broadcast angle for distinguishing between standard fouls and bookable offenses.

Performance Analysis. The ensemble model demonstrates varying degrees of success across the task heads, as summarized in Figure 7, highlighting both improvements in specific areas and persistent challenges with class imbalance.

As shown in Table 11, the offence prediction head achieved an overall accuracy of 92.67%, whereas the Balanced Accuracy remains at 50.00%. The model completely collapsed into the majority class, predicting "Offence" for all samples and failing to identify any of the "No Offence" instances (0.00% recall). This extreme mode collapse is not solely a byproduct of numerical imbalance, but also reflects the intrinsic visual ambiguity of the domain. As noted by the dataset's original authors [14], the visual boundary between a clean defensive intervention ("No Offence") and a minor infraction is exceptionally subtle. The distinction often relies on micro-dynamics, such as the exact point of contact, the player's intention, or the speed of the challenge, that manifest only in a handful of frames. Unable to consistently extract these highly granular and transient features, the network defaults to the statistically safer majority prediction.

This complete failure on the "No Offence" class should be interpreted as one of the most informative findings of the study, as it suggests that the visual boundary between a legal challenge and a minor infringement is often encoded in highly localized micro-interactions that remain effectively invisible once frame-level representations are globally pooled and then temporally compressed through recurrent aggregation, making the collapse not simply an optimization artefact but evidence of a representational bottleneck.

The Action Classification head achieved an overall accuracy of 32.00% and a Balanced Accuracy of 17.83%. Notably, the model showed distinct capabilities in identifying specific lower-body physical interventions, achieving 50.00% accuracy on "Tackling", 46.03% on "Standing tackling", and 44.44% on "High leg". However, it completely failed to recognize other upper-body or less frequent classes such as "Holding", "Pushing", and "Elbowing" (all 0.00%). This discrepancy underscores the varying visual complexity of different soccer fouls. While lower-body interventions typically exhibit pronounced and easily recognizable kinematic trajectories, illegal uses of the arms share highly overlapping physical movements and frequently occur in crowded areas with heavy occlusion, making them difficult to disentangle. Furthermore, highly contextual or deceptive actions, such as "Dive" or generic "Challenge" situations, do not exhibit uniquely generalizable spatiotemporal signatures, which further increases the difficulty of the classification task. Nevertheless, our model still achieves a recall of approximately 20% on this category, suggesting that it is able to capture at least part of the discriminative signal despite the strong visual ambiguity.

Table 11. Ensemble pipeline performance metrics.

Task Head	Accuracy	Balanced Accuracy
Severity	61.33%	**46.87%**
Offence	92.67%	50.00%
Action	32.00%	17.83%

The hybrid cascading logic for the Severity assessment yielded an overall accuracy of 61.33% and a Balanced Accuracy of 46.87% detailed in Table 12. As shown in Figure 7, the ensemble successfully recovered a portion of the minority class, correctly identifying 1 out of 5 "Red Card" instances, such as in the second pipeline. Furthermore, it showed consistent performance on the majority classes, identifying "No Card" at 63.68% and "Yellow Card" at 56.94%.

Table 12. Ensemble: detailed confusion metrics for severity head.

Severity Class	Correct / Total	Class Recall
No Card	142 / 223	63.68%
Yellow Card	41 / 72	56.94%
Red Card	1 / 5	20.00%
Overall Metrics	**Acc:** 61.33%	**Bal. Acc:** 46.87%

† One No Card instance was skipped during inference due to a batch-boundary drop in the test loader.

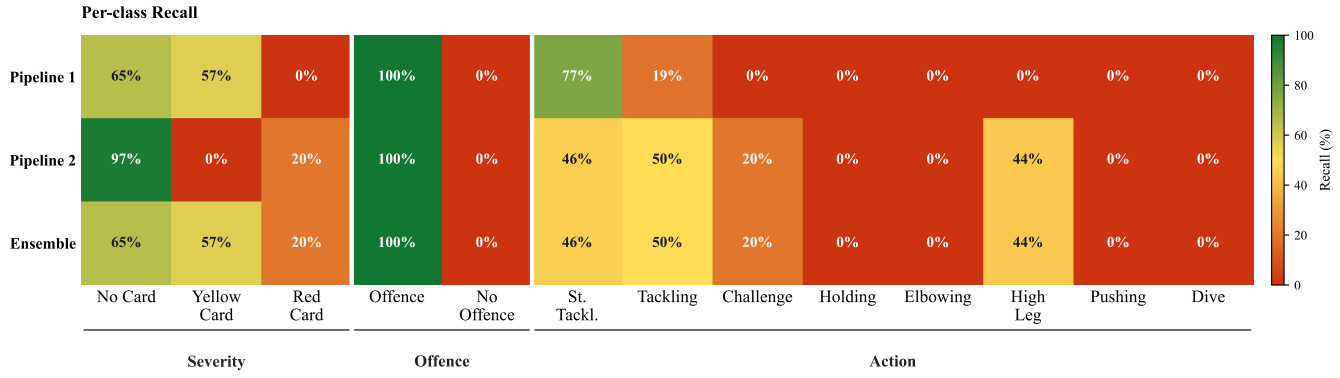

Figure 7. Per-class recall across all primary task heads.

6.5 Comparison with Prior Refereeing Systems

To contextualize the performance of ArbItro, it is essential to compare our results with prior refereeing-oriented systems developed on SoccerNet-MVFouls, starting from the reference formulation introduced by Held et al. [14].

Task Formulation and Architectural Divergence. A direct one-to-one numerical comparison with prior VARS-based systems remains inherently difficult because the underlying task formulations are not fully aligned. The original VARS architecture models the disciplinary outcome as a unified 4-class prediction problem ("No offence", "Offence + No card", "Offence + Yellow card", and "Offence + Red card"), whereas ArbItro decomposes the decision process into two separate but related outputs: a binary *Offence* prediction and an independent 3-class *Severity* assessment. This design choice is intended to disentangle the visual recognition of an infringement from the subsequent estimation of its disciplinary consequence.

In addition, the compared baseline relies on dedicated spatiotemporal encoders. While modern video understanding increasingly relies on 3D convolutional architectures such as SlowFast [11] and transformer-based spatiotemporal models such as TimeSformer [1] and VideoMAE [31], these approaches typically operate in a substantially richer representational regime than lightweight CNN–RNN pipelines. Accordingly, the comparison is best interpreted not as a direct head-to-head competition, but as an analysis of how different points in the design space differ in representational capacity, task formulation, and robustness under imbalance and visual ambiguity. Consequently, ArbItro adopts a heterogeneous cascading design built around a single-clip CNN–BiLSTM branch and a multi-clip masked-mean branch, not as a claim of modern efficiency, but as a controlled study of what a structurally classical pipeline can recover on this benchmark. Rather than claiming a uniformly stronger architecture, our results suggest that the two branches exhibit complementary behavior: the single-clip pathway appears comparatively more stable on frequent and visually simpler cases, whereas the multi-clip pathway provides limited but non-zero sensitivity to rarer and more ambiguous events. This distinction should nonetheless be interpreted cautiously, as the dataset remains relatively small and strongly imbalanced, especially for high-severity outcomes.

Performance and the Minority Class Bottleneck. Both frameworks expose the extreme difficulty of overcoming the dataset's long-tail distribution and visual ambiguity. Notably, when utilizing a CNN-based feature extractor (ResNet) comparable to our spatial backbone, the original baseline achieved a Balanced Accuracy of only 25% on the multi-view severity task. Even when upgrading to their best multi-task Vision Transformer (MViT) across all available camera views, the baseline only reached a Balanced Accuracy of 39% on the combined 4-class offence and severity classification. A deeper look at their confusion matrices reveals that these advanced attention mechanisms struggled heavily with minority classes: their model achieved a 14% precision when recognizing "No offence" events, and while it reached a 27% recall on "Red Card" offences, this came at the cost of a catastrophic 5% precision, indicating massive over-prediction and confusion with neighboring classes. Compared to these transformer-based and fully multi-view architectures, ArbItro deliberately relies on a structurally classical 2D CNN + BiLSTM design and a deterministic ensemble rather than on heavy attention-based spatiotemporal encoders. The fact that both approaches still struggle with the same minority classes suggests that dataset imbalance, occlusions, and label ambiguity remain dominant bottlenecks, and that improvements in model capacity alone are unlikely to resolve the representational difficulty of the foul boundary.

Our ensemble mirrors this struggle on the "No Offence" boundary, exhibiting a complete mode collapse (0% recall) on the binary Offence head. However, under our decoupled formulation, the isolated Severity head within ArbItro's cascading logic achieved a Balanced Accuracy of 46.87%. Crucially, under this decoupled 3-class severity formulation, the

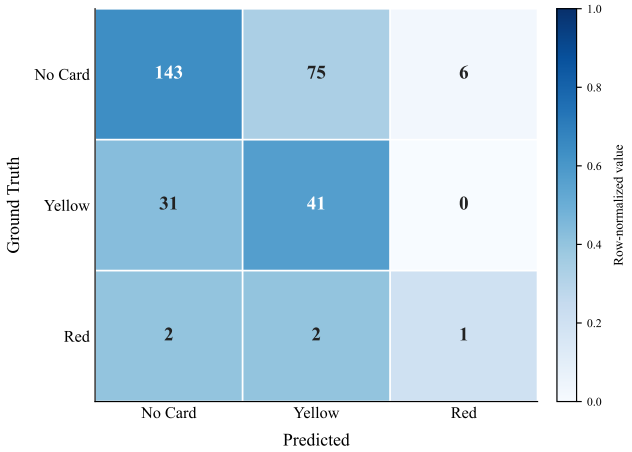

Figure 8. Confusion matrix of the ensemble on the 3-class Severity head.

ensemble maintained a 20% recall on the extremely rare "Red Card" events with a corresponding precision of 14%, avoiding the severe precision degradation reported by the baseline in its joint 4-class setting. At the same time, the model effectively stabilized the prediction of standard fouls, achieving 63.68% and 56.94% recall for "No Card" and "Yellow Card", respectively. This class-wise prediction pattern is further illustrated by the confusion matrix in Figure 8.

Current Landscape. Since the release of the SoccerNet-MVFouls dataset, the landscape of automated refereeing has rapidly shifted towards multi-modal and generative approaches. Recent state-of-the-art frameworks, such as X-VARS [16], have integrated Multi-Modal Large Language Models (LLMs) to process visual encoder embeddings, substituting pure kinematic classification with deep semantic reasoning and language-driven explainability. At the same time, pure classification performance continues to scale through heavier spatial architectures. As officially reported in the recent SoccerNet 2025 Challenges [13], the new state-of-the-art was set at 52.22% of combined Balanced Accuracy. Notably, this peak performance was achieved by employing large Vision Transformers (ViT-L) coupled with adaptive multi-view fusion, whereas LLM-based systems reached a highly competitive 49.32% through zero-shot structured prompting.

Within this context, ArbItro should not be interpreted as a direct competitor to the strongest contemporary multimodal systems in terms of absolute performance. Its contribution lies instead in showing that a structurally classical CNN–RNN pipeline can still recover part of the disciplinary signal, while making explicit where this family of models breaks down under imbalance, occlusion, and ambiguous contact dynamics. More broadly, the comparison highlights that the central limitation of football foul recognition is not only architectural scale, but the difficulty of preserving the localized and short-duration evidence required to resolve the foul boundary. In this sense, ArbItro is most useful not as an efficiency-aware baseline, but as an empirical characterization of which aspects of the task remain inaccessible without richer spatiotemporal attention, cross-view alignment, or multimodal reasoning.

7 Conclusion

This paper presented ArbItro, a multi-task CNN–RNN framework for football foul recognition on SoccerNet-MVFouls, evaluated as a controlled empirical study of what structurally classical architectures can recover under severe class imbalance, occlusion, and fine-grained visual ambiguity.

The empirical results show that the two pipelines exhibit partially complementary behavior. The single-clip branch remains comparatively more stable on frequent disciplinary outcomes, whereas the multi-clip branch provides limited but non-zero sensitivity to rarer and more severe events. At the same time, the overall evaluation exposes clear structural limitations: the ensemble reaches a Balanced Accuracy of 46.87% on the 3-class Severity task, but still collapses completely on the binary Offence head, failing to recover the "No Offence" class.

These findings suggest that, within structurally classical frameworks, the main bottlenecks of football foul recognition are not solely optimization-related, but are strongly tied to the representational limits of CNN–RNN pipelines when subtle contact dynamics must be inferred from broadcast footage. In this sense, ArbItro should not be interpreted as a production-ready decision-support system, nor as an efficiency-oriented baseline, but as an empirical characterization of where this architectural family remains insufficient.

More broadly, our results indicate that task decoupling and deterministic cascading can partially stabilize disciplinary prediction, but they are insufficient to resolve the intrinsic ambiguity of minority classes and foul boundaries. The contribution of this work therefore lies less in establishing a new state of the art and more in clarifying which aspects of the task require richer spatiotemporal and multimodal representations.

7.1 Future Work

Despite the architectural refinements explored in this work, the intrinsic visual ambiguity of football foul recognition remains a formidable challenge. The boundary between a legal physical contest and a bookable foul is often dictated by high-frequency micro-dynamics, such as sub-frame variations in postural alignment, points of bodily contact, and impact momentum, that unfold in windows of less than 500 milliseconds. Resolving these instances requires preserving localized spatiotemporal evidence that often eludes standard 2D+1D temporal aggregation schemes [33]. Furthermore,

severe infractions frequently occur within crowded penalty areas characterized by strong inter-object occlusion and extreme player density. Under these conditions, standard broadcast views often fail to preserve the discriminative signal, reinforcing the need for richer multi-view spatiotemporal modeling [14].

Future work should therefore investigate more expressive video architectures, including masked video autoencoders [31] and recent Mamba-based video models [20, 23, 24], which are promising not only for long-range dependency modeling, but also for their potential to retain sparse and highly localized evidence across time. In particular, the limits observed in our CNN–BiLSTM pipelines suggest that future progress may depend less on additional task-specific tuning and more on adopting model families that can preserve short-duration contact cues without compressing them too aggressively through global pooling and recurrent aggregation.

Beyond architectural improvements, an equally important direction concerns the evolution of the benchmark itself. Given the strong class imbalance and the limited support for rare disciplinary outcomes, future work should investigate curated dataset expansion strategies aimed at increasing the coverage of minority events, borderline contacts, and visually ambiguous "No Offence" scenarios. Such an effort would not only improve class balance, but also help align the benchmark with the evolving standards of modern refereeing, where disciplinary interpretation is continually reshaped by player-safety guidelines and VAR-mediated review protocols. In parallel, expanding the dataset with ultra-slow-motion material, referee-centric viewpoints, and more recent match footage may help bridge the current gap between machine prediction and expert human arbitration.

From an architectural standpoint, future iterations should also investigate the integration of Vision-Language Models (VLMs) and Multi-Modal Large Language Models (LLMs), in line with recent developments in explainable refereeing systems. While our current deterministic ensemble offers a structurally classical pipeline for kinematic foul analysis, coupling it with generative semantic reasoning may improve both interpretability and robustness, especially in borderline incidents where visual evidence alone remains insufficient. Additionally, self-supervised pre-training and contrastive learning paradigms may provide stronger foundational representations, reducing the framework's dependence on heavily skewed manual annotations. More broadly, the next step for this line of research lies not only in building stronger models, but in combining richer representations with more representative supervision.

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